

AMEY SHUKLA

PERSONAL INFORMATION

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GOAL

Advance practical cryptography by developing theoretical tools for authentication and privacy that address challenges in building resilient systems. Enable non-experts to implement resilient systems effectively.

EDUCATION

<i>PhD in Computer Science</i>	<i>2023–2028</i> University of Connecticut In the area of applied cryptography under Dr. Benjamin FULLER.
<i>Masters in Computer Science, Thesis</i>	<i>2021–2023</i> Wichita State University In the area of cybersecurity under Dr. Sergio Salinas MONROY.
<i>Bachelors in Computer Science</i>	<i>2017–2021</i> Wichita State University Dual/Accelerated program under Dr. Sergio Salinas MONROY.

WORK EXPERIENCE

<i>University of Connecticut</i>	<i>2023–Present</i> Graduate Research Assistant Working on exciting problems in applied cryptography, such as noisy authentication, oblivious computation, and searchable encryption.
<i>Persona Identities</i>	<i>Summer 2026</i> Research Scientist, Intern Worked on privacy-first identity verification; specifically, how we can store and compare identities without ever saving raw biometrics. Writing new research on new & exciting fuzzy extractors and their applications to fraud & identity.
<i>Wichita State University</i>	<i>2021–2023</i> Graduate Teaching Assistant TA'd: Introduction to Cybersecurity, Routing & Switching I, Applied Computing & Networks, Applied Programming & Scripting, Applied Web Apps & Database Development, as well as designed & taught a brand new CTF prep course from scratch (now a required course for applied computing cybersecurity track degree). Responsibilities: Taught material, graded assignments, reached out and guided struggling students, reinforced material, answered student questions during office hours, and tracked attendance.
<i>NetApp</i>	<i>2019–2021</i> Software Engineer Intern NetApp, formerly Network Appliances, is a storage appliance company that focuses on SAN, NAS, and other hybrid-cloud storage solutions. My work was mostly written in Java, Python, and C++. Developed and maintained E-Series REST API, which served as an interface to low-level details of the storage appliance.

Developed a next-generation container execution framework utilizing Kubernetes, Docker, and Ansible to run storage software and functions on corresponding software with load balancing and highly available topology.

Developed a resource provisioning framework to provision storage software builds virtually on demand, saving many hours of developer downtime.

Implemented stress tests for storage appliances to test recovery in case of faults.

PUBLICATIONS

Note: Many are alphabetical by last name, as is the convention for cryptography research.

Eprint Noah Arce-Caliskan, Benjamin Fuller, Gadalia Montoya Weinberg O'Bryan, Maryam Rezapour, Amey Shukla.
Sample-then-Correct: Practical fuzzy extractors for biometrics beyond the iris. ePrint.

Eprint Benjamin Fuller, Ariel Hamlin, Arinjita Paul, Maryam Rezapour, Ronak Sahu, Amey Shukla, Mason Stuart.
Updatable Oblivious Key Value Stores with Access Control and Application to Multi Key Searchable Encryption. ePrint.

Eprint Mahdiah Heidari pour, Sage Pia, Maryam Rezapour, Amey Shukla, Gagan Agrawal, Hoda Maleki, Benjamin Fuller.
Block Oblivious Single Access Machines and Oblivious Graph-Based Approximate Nearest Neighbor Search with Small Client Storage. ePrint.

Eprint Sage Pia, Ananya Appan, Maryam Rezapour, Amey Shukla, Nikhil Date, Benjamin Fuller, Ling Ren, David Heath.
Oblivious Single Access Machines are Concretely Efficient. ePrint.

ACM CCS Amey Shukla, Luke Demarest, Benjamin Fuller, Sohaib Ahmad, Caleb Manicke, Alexander Russell, Sixia Chen.
Fuzzy Extractors are Practical: Cryptographic Strength Key Derivation from the Iris. ACM CCS 2025.

Eprint Benjamin Fuller, Arinjita Paul, Maryam Rezapour, Ronak Sahu, Amey Shukla.
MARS: Low-Leakage Multi Adversarial Owner and Reader Replication-free Searchable Encryption from Private Information Retrieval. ePrint.

OTHER INFORMATION

Awards 2026 · UConn Computer Science Predoctoral Fellowship.
2025 · ACM CCS Travel Award.
2025 · UConn Computer Science Predoctoral Fellowship.
2025 · UConn Fall Conference Participation Award.
2023 · UConn CSE Fellowship.
2021 · Global Select Scholarship.

Invited Talks **Sample then Correct.**

New England Security Day 2026 · Talk about amplification for fuzzy extractors at the University of Massachusetts Amherst.

Fuzzy Extractors are Practical.

Google · Talk about Fuzzy Extractors at Google.

Microsoft Research · Talk about Fuzzy Extractors at Microsoft Research.

JP Morgan · Talk about Fuzzy Extractors at JP Morgan.

Boston University · Talk about Fuzzy Extractors at BU Crypto Seminar.

Persona · Talk about Fuzzy Extractors at Persona.

OTHER EXPERIENCES

Peer Review

Subreviewer: IEEE S&P 2026, USENIX Security 2026, ACM CCS 2024.

Kansas Science Olympiad

Invited to design the Codebusters examination for Kansas Science Olympiad.

Wichita State Cybersecurity Bootcamp

Invited to help moderate a cybersecurity summer camp for children in Middle & High School. Helped children during workshop, answered questions, debugged their code, and tasks to point out issues.

ACE Mentor at Wichita State

Volunteered at a program that helped university freshman make the best of their first semester.

2019–2021

Vice President

Wichita State Linux Users Group

Managed a Linux Users Group with 100+ members, and led workshops on applied cryptography, security, systems programming, ansible, computer graphics, and of course Linux.

Keywords

Rust · C · Python · Git · L^AT_EX · Docker · Ansible · Kubernetes · CI/CD · KVM · Mercurial · Cryptography.

Can write most imperative programming languages with access to documentation.

September 16, 2026